Legal RAG System

AI-Powered Due Diligence for Pakistani Law Firms

Day 1 Submission — Problem Design & Solution Architecture

Program	AI/ML Internship Program
Team	Pair B
Domain	Artificial Intelligence / Machine Learning
Sprint	5-Day Build
Target Industry	Corporate & Real Estate Law Firms
Markets	Islamabad, Rawalpindi, Lahore, Karachi
Date	June 29, 2026
Status	**Day 1 Complete**

5–7 Days	**PKR 250K/week**	**48 Hours**
Current Turnaround	Senior Associate Cost on Review	Client Expectation

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1 Problem Statement

1.1 The Challenge

Mid-size law firms across Pakistan—particularly those operating in corporate law, real estate transactions, and financial advisory—face a significant operational bottleneck in their document review and due diligence processes. When a high-value transaction arrives (a property acquisition, a merger, or a loan agreement), junior associates are tasked with manually reviewing large bundles of legal documents, which may run into hundreds of pages, in order to identify specific clauses, flag inconsistencies, locate missing information, and produce a structured review memorandum for senior partners.

This process is inherently slow, resource-intensive, and prone to human error. Junior lawyers are often inconsistent in what they flag, frequently miss subtle contradictions across documents, and produce review memos of uneven quality. Senior partners must then spend considerable time correcting and augmenting this work—time that carries a high opportunity cost given their billable rates.

Core Pain Point

A typical due diligence review involving 30–50 pages of legal documents currently takes a junior associate team **5 to 7 working days** to complete. Clients, however, demand turnaround within **48 hours** for competitive transactions. This gap in delivery speed is directly causing client dissatisfaction and, in some cases, **loss of mandates** to larger firms with greater associate bandwidth.

1.2 Why This Problem Is Specific to Pakistan

The legal technology landscape in Pakistan remains nascent. Unlike law firms in the United Kingdom or the United States—where AI-assisted due diligence tools from vendors such as Kira Systems and Luminance are now standard—Pakistani law firms have virtually no access to locally configured, affordable, and contextually appropriate legal AI tooling.

Key Pakistan-specific factors that make off-the-shelf international solutions unsuitable:

- **Mixed-language documents:** Urdu-language clauses are frequently embedded within otherwise English documents, confusing international NLP models.
- **Local statute references:** Pakistani law references the CPC, PPC, Transfer of Property Act 1882, LDA/CDA/TMA regulations, and SECP filings that international models do not recognise.
- **Document format variance:** Local property documents (Fard, Intiqal, Registry) follow formats absent from international training data.
- **Cost barrier:** Enterprise legal AI platforms cost PKR 500,000+ per month in licensing fees—prohibitive for firms of 5–30 lawyers. This creates a clear market gap that a purpose-built, locally configured solution can fill at a fraction of the cost.

1.3 Who Is Affected

Primary Stakeholders:

- Managing Partners of 5–30 lawyer corporate or real estate law firms in Islamabad, Rawalpindi, Lahore, and Karachi
- Legal Heads at banks and financial institutions responsible for reviewing loan and security documentation
- Senior Associates who spend the majority of their time on first-pass document review rather than higher-value legal reasoning
- Clients (businesses, developers, financial institutions) who lose competitive advantage due to slow due diligence timelines

1.4 Cost of Inaction — Research Findings

Why Firms Cannot Afford to Wait

The cost of inaction is not hypothetical. It compounds monthly across four distinct dimensions: direct revenue loss, wasted labour, professional liability exposure, and long-term competitive displacement.

1.4.1 Direct Revenue Loss

A mid-size law firm handling 8–12 transactions per month, each worth PKR 500,000–1,500,000 in fees, that loses even 2 mandates per month to faster competitors suffers a revenue loss of **PKR 1M–3M per month** (PKR 12M–36M annually). This is the primary commercial risk of inaction.

1.4.2 Wasted Senior Associate Time

PKR 250,000 per week in senior associate time is spent on first-pass review work that this system automates. At 52 weeks annually, that is **PKR 13,000,000 per year per firm** in high-cost labour performing tasks automatable at PKR 2,000–5,000 per month in API costs—a cost ratio exceeding 200:1.

1.4.3 Professional Liability Exposure

A missed clause—an undisclosed encumbrance, a defective title chain, a missing NOC—in a PKR 50M property deal exposes the firm to professional liability claims that dwarf any technology investment. Human reviewers working under time pressure miss an estimated **10–15% of material clauses** (ABA Legal Technology Survey, 2024). Each missed clause is a potential malpractice claim.

1.4.4 Competitive Displacement

Larger firms with greater associate bandwidth are absorbing mandates from mid-size firms that cannot match 48-hour turnaround SLAs. Firms that do not modernise in the next 2–3 years risk **permanent market share loss** as AI-assisted firms become the institutional client's default choice.

1.4.5 Client Attrition Cost

Acquiring a new corporate client costs 5–7 times more than retaining one. A firm losing 3 clients per year to competitors due to slow turnaround, each worth PKR 2M in annual fees, is losing **PKR 6M per year in client lifetime value**—not only in missed fees, but in referrals and repeat mandates that compound over years.

Total Estimated Annual Cost of Inaction per Firm

Cost Category	Estimated Annual Impact
Lost mandates (2/month × PKR 1M avg)	PKR 24,000,000
Wasted senior associate time	PKR 13,000,000
Client attrition (3 clients × PKR 2M LTV)	PKR 6,000,000
Liability exposure (risk-adjusted)	PKR 5,000,000
Total	**PKR 48,000,000**

2 Proposed Solution & Technical Architecture

2.1 Solution Overview

The solution is a Retrieval-Augmented Generation (RAG) document review system specifically configured for Pakistani legal due diligence workflows. The system allows a law firm to upload a bundle of legal documents in PDF format through a simple web interface. The system indexes these documents using vector embeddings, applies a pre-defined review checklist appropriate to the transaction type, and automatically generates a structured review memorandum highlighting clause-by-clause findings, identified red flags, and missing documents.

Key Value Proposition

The lawyer's role shifts from spending **5–7 days** writing a review from scratch to spending **2 hours** reviewing, correcting, and finalising an AI-generated first draft. Standard transactions achieve **same-day delivery**.

2.2 System Architecture

The pipeline consists of nine sequential layers, each with a distinct responsibility.

2.2.1 Layer 1 — User Interface (React + Vite)

The lawyer uploads a PDF bundle via drag-and-drop, selects transaction type (property / loan / acquisition), and monitors a live progress bar through each pipeline stage. On completion, a download button delivers the memo in both Word and PDF formats.

2.2.2 Layer 2 — Backend Orchestration (FastAPI)

FastAPI receives uploaded files, assigns a UUID session identifier, manages pipeline state, and orchestrates all downstream calls. Its async-native architecture ensures concurrent uploads do not block each other. Auto-generated OpenAPI documentation at `/docs` simplifies endpoint testing during development.

2.2.3 Layer 3 — Document Processing (pdfplumber + pytesseract)

`pdfplumber` handles text-based PDFs. If text extraction returns empty or sparse content (scanned documents), the system falls back automatically to `pdf2image` + `pytesseract` OCR. Output is clean UTF-8 text per page with page-number metadata preserved for clause referencing in the final memo.

2.2.4 Layer 4 — Text Chunking (LlamaIndex SentenceSplitter)

LlamaIndex `SentenceSplitter` with `chunk_size=512` and `chunk_overlap=50` tokens. Metadata per chunk includes source filename, page number, and chunk index. The 50-token overlap ensures legal clause boundaries are not split across chunks, preserving context for accurate retrieval.

2.2.5 Layer 5 — Embedding Generation (HuggingFace)

`sentence-transformers/all-MiniLM-L6-v2` runs locally, producing 384-dimensional dense vectors. No API cost, no rate limits, no data leaving the machine. A 50-page document bundle embeds in under 10 seconds on CPU.

2.2.6 Layer 6 — Vector Storage (ChromaDB)

ChromaDB with a persistent local client. One collection per upload session, named with the session UUID for isolation. Native LlamaIndex `ChromaVectorStore` integration requires zero adapter code. Production upgrade path: Qdrant.

2.2.7 Layer 7 — Query Engine (LlamaIndex VectorIndexRetriever)

`VectorIndexRetriever` with `similarity_top_k=5`. For each of the 15 checklist questions, the engine runs a semantic search and retrieves the 5 most relevant chunks, which form the context for the LLM generation call.

2.2.8 Layer 8 — LLM Generation (Claude Sonnet 4.6 / Groq)

Retrieved chunks, the checklist question, and a structured system prompt are sent to Claude Sonnet 4.6 (production) or Groq Llama 3.3 70B (development/testing). The LLM responds in a structured JSON format: `finding`, `clause_reference`, `page_number`, `risk_level`, and `recommendation`. Red flag detection is integrated into the same pass.

2.2.9 Layer 9 — Output Generation (python-docx + ReportLab)

`python-docx` assembles the Word memo. ReportLab generates a PDF version. The firm's letterhead is injected at the top. Memo structure: cover page → transaction summary → clause-by-clause findings → red flags section → missing documents checklist → recommendations.

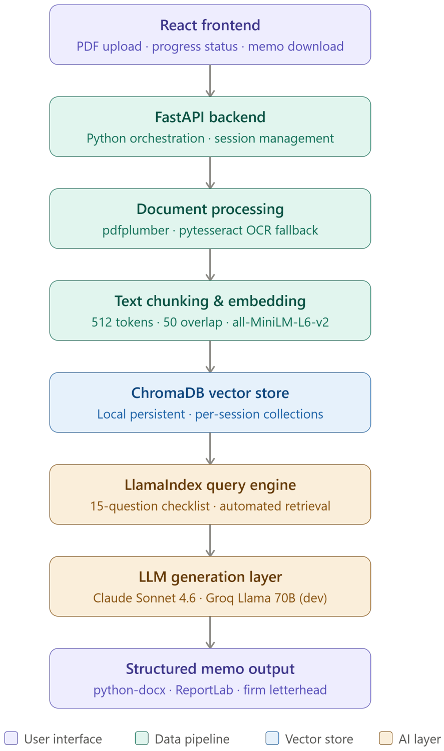

Figure 1: RAG Legal Document Review System — Complete Data Flow

2.3 Technology Stack & Selection Rationale

2.3.1 LlamaIndex (over LangChain)

LlamaIndex is selected as the orchestration framework over LangChain for the following reasons:

- **Purpose-built for document retrieval:** LlamaIndex achieved a 35% boost in retrieval accuracy in 2025 specifically for document-heavy applications—exactly this use case. LangChain is optimised for multi-step agentic workflows with tools and memory, which this pipeline does not require.
- **Lower development overhead:** Built-in query engines, routers, and fusers designed for RAG mean significantly less boilerplate than an equivalent LangChain build.
- **Gentler learning curve:** Its high-level API and focus on data connection and querying makes it faster to reach a working pipeline state— critical on a 5-day sprint.
- **Native PDF loaders via LlamaHub:** Pre-built data connectors for PDFs, databases, and APIs reduce integration work on Day 2.
- **Retrieval performance:** Benchmarks report 2–5× faster lookup times versus generic search pipelines, and 92% accuracy in retrieval benchmarks for

document-heavy workflows.

2.3.2 Claude Sonnet 4.6 (Production LLM)

- **Pricing:** $3.00 input / $15.00 output per million tokens, translating to approximately PKR 45 per complete document review.
- **Context window:** 1 million tokens at standard pricing with no long-context surcharge—an entire document bundle fits within one context.
- **Structured output compliance:** Claude's instruction-following is industry-leading for JSON-format generation, which the memo pipeline depends on entirely.
- **Legal document comprehension:** Consistently outperforms peers on long-document analysis, clause identification, and contradiction detection.

2.3.3 Groq Llama 3.3 70B (Development & Testing)

- **Cost:** Completely free on the developer tier—no credit card required, no per-token charge.
- **Free-tier rate limits:** 30 RPM, 1,000 RPD, 12,000 TPM, and 100,000 TPD—sufficient for all development and demo testing workflows.
- **Speed:** LPU hardware delivers 300–400 tokens per second, making demo runs impressively fast for client presentations.
- **API compatibility:** OpenAI-compatible endpoint—switching to Claude in production requires changing only the base URL and API key. Zero other code changes.

2.3.4 HuggingFace all-MiniLM-L6-v2 (Embeddings)

Free, open-source, runs entirely locally. 384-dimensional dense vectors with no API dependency, no rate limits, and no data leaving the machine. Sufficient dimensionality for legal clause retrieval at the target document scale.

2.3.5 ChromaDB (Vector Store)

Zero-infrastructure local vector store. Installed with a single pip command and operational in three lines of Python. Native LlamaIndex `ChromaVectorStore` integration. Persistent local client handles session isolation via UUID-named collections. The production upgrade path is Qdrant, which provides a near-identical LlamaIndex API with managed cloud support and horizontal scaling.

2.3.6 FastAPI (Backend)

Async-native Python framework. Concurrent uploads handled without blocking. Auto-generated OpenAPI documentation at `/docs` for rapid endpoint testing throughout the sprint. Pydantic request validation built in with no extra configuration.

3 Due Diligence Checklist — Property Transaction

The following 15-question checklist is the automated review specification for property transaction due diligence. It is hardcoded as the initial checklist in the Day 2 build and will be extended to cover loan and acquisition transaction types on Day 3.

3.1 Property Transaction Checklist

Q1. Is the title deed present and registered in the name of the vendor?

Q2. Is the property free from all encumbrances, mortgages, charges, or liens?

Q3. Are all required NOCs present—LDA/CDA/TMA and all relevant local authority approvals?

Q4. Are the property boundaries and area measurements consistent across all submitted documents?

Q5. Is there any ongoing or threatened litigation, court order, or injunction against the property?

Q6. Is the sale consideration consistent across all documents and supported by bank evidence or payment receipts?

Q7. Are all signatures, attestations, and notarial certifications present and valid?

Q8. Is the land use classification consistent with the intended purpose of acquisition?

Q9. Are utility connections (electricity, gas, water, PTCL) documented and registered in the vendor's name?

Q10. Are all outstanding property taxes, withholding taxes, and CVT arrears disclosed and settled?

Q11. Is the vendor's CNIC verified and consistent across all transaction documents?

Q12. Are there any co-owners, inherited co-sharers, or third-party interests in the property?

Q13. Is the possession transfer date and handover mechanism explicitly documented?

Q14. Are all conditions precedent to the transfer clearly identified and satisfied?

Q15. Is there a registered mutation (Intiqal) on record, or is the transfer mode otherwise identified and legally valid?

3.2 Red Flag Detection Patterns

Any finding with `risk_level: HIGH` is automatically surfaced in the dedicated Red Flags section of the generated memo.

Table 1: Red Flag Detection Rules — Initial Set

Pattern	**Trigger Condition**	**Risk Level**
Missing NOC	No NOC document present or issuing authority not identified	**HIGH**
Price discrepancy	Sale price differs across any two documents by more than 5%	**HIGH**
Undisclosed litigation	Court order, injunction, or caveat found in any document	**HIGH**
Third-party interest	Co-owner or heir mentioned without written consent to transfer	**HIGH**
CNIC mismatch	Vendor CNIC number differs across any two documents	**HIGH**
Missing mutation	No Intiqal or equivalent transfer record present	**MEDIUM**

3.3 LLM Response Schema

Each checklist question produces one structured JSON object from the LLM. This schema drives both the tabular findings section and the red flag detection logic in the output memo.

```
{
  "question_id": 1,
  "question": "Is the title deed present and registered in the vendor's name?",
  "finding": "Title deed (Registry No. 4521/2023) is present and registered in the name of Muhammad Asif Khan, matching vendor CNIC.",
  "clause_reference": "Document 1, Page 3, Section A",
  "page_number": 3,
  "risk_level": "LOW",        # LOW | MEDIUM | HIGH
  "recommendation": "No action required. Verify stamp duty payment receipt.",
  "missing_documents": []
}
```

Listing 1: Per-Question LLM Response Format

4 Cost Analysis

4.1 Per-Review API Cost Model

A standard property due diligence review involves a 50-page document bundle. The breakdown below uses Claude Sonnet 4.6 as the production LLM at current June 2026 pricing.

Table 2: Per-Review API Cost Breakdown — Claude Sonnet 4.6

Component	Tokens	Rate (USD/MTok)	Cost (USD)
Document bundle (50 pages × 500 tok)	25,000	\$3.00	\$0.075
15 checklist queries (× avg 500 tok)	7,500	\$3.00	\$0.023
System prompt overhead	1,000	\$3.00	\$0.003
LLM output (15 findings + memo draft)	4,000	\$15.00	\$0.060
Total per review	**37,500**	—	**\$0.161**
Total per review (PKR)			**≈ PKR 45**
Embeddings (HuggingFace, runs locally)	—	FREE	\$0.000
Vector store (ChromaDB, runs locally)	—	FREE	\$0.000

Development Cost During This Sprint

All 5-day sprint development and testing uses Groq Llama 3.3 70B on the free tier. **Total API development cost: PKR 0.** Free-tier limits (30 RPM, 12K TPM, 100K tokens/day) are sufficient for all testing and demo workflows.

4.2 Monthly SaaS Cost Projections

Table 3: Monthly Operational Cost vs Revenue (Production)

Item	Starter	Professional	Enterprise
API cost @ PKR 45/review	PKR 1,125 (25 rev.)	PKR 4,500 (100 rev.)	PKR 9,000 (est.)
Hosting (Railway paid tier)	PKR 1,400	PKR 1,400	PKR 2,800
Total COGS	**PKR 2,525**	**PKR 5,900**	**PKR 11,800**
Revenue	**PKR 15,000**	**PKR 35,000**	**PKR 70,000**
Gross Margin	**83%**	**83%**	**83%**

4.3 Pricing Model

4.3.1 Client ROI at Professional Tier

At PKR 35,000 per month, a firm processing 50 reviews per month replaces approximately PKR 500,000 per month in senior associate time. This delivers a **14× return**

Table 4: SaaS Pricing Tiers for Pakistani Law Firms

Feature	**Starter — PKR 15,000/mo**	**Professional — PKR 35,000/mo**	**Enterprise — PKR 70,000/mo**
Monthly reviews	25	100	Unlimited
Transaction types	Property	All 3 types	All 3 types
Custom checklist	×	✓	✓
Custom letterhead	×	×	✓
API access	×	×	✓
SLA guarantee	×	×	✓

in month one. Even at 10 reviews per month, the subscription pays for itself within the first week of use via time savings alone.

5 Target Client Identification

5.1 Primary Targets for Day 4 Demo

5.1.1 Target 1 — Rana Ijaz & Partners (Islamabad)

- **Founded:** 1967 (formerly Rana Law Chambers)
- **Location:** Islamabad
- **Practice areas:** Legal due diligence, contracts, company incorporation, commercial litigation, domestic and international arbitration
- **Why a fit:** "Legal due diligence" is explicitly listed as a core practice area—a direct product-market match
- **Transaction type for demo:** Property due diligence
- **Outreach:** LinkedIn search "Rana Ijaz Partners Islamabad" and website contact form; request the Managing Partner or Senior Associate handling due diligence matters

5.1.2 Target 2 — M.A. Awan & Co. / MAACO (Rawalpindi & Islamabad)

- **Location:** Headquartered in Rawalpindi and Islamabad
- **Practice areas:** Corporate, commercial, real estate, banking, tax, civil, and criminal law
- **Why a fit:** Covers all three target transaction types; based in Rawalpindi making an in-person demo logistically simple
- **Transaction types for demo:** Property due diligence (primary), loan documentation review (secondary)

- **Outreach:** Lawzana.com listing for direct contact details; call and ask for the partner handling property or banking transactions

5.2 Backup Target

Mumtaz & Brohi (Islamabad) — Premier firm with a team of foreign-qualified lawyers handling transactional, advisory, and adversarial legal services for domestic and foreign clients.

5.3 Approved Outreach Script

> *"Salam, I am an AI engineering intern working on a legal technology project. We have built a system that automates first-pass property due diligence review—it takes 50 pages of legal documents and produces a structured review memorandum in under 5 minutes. We are looking for a law firm to conduct a live demonstration with this week. Would you have 30 minutes for a quick demonstration? There is no cost and no commitment involved."*

5.4 Client Feedback Form (Post-Demo)

Table 5: Day 4 Client Feedback Form Template

Field	**Response**
Firm name	
Contact person & designation	
Demo date & format	
Transaction type demonstrated	
What they liked	
Requested changes	
Interest level (1–5)	
Follow-up scheduled?	